Data Availability Statement

Manuscript: Symmetric Molecular Crystal Flow Matching with Pair-Bias Attention and Rigid-Body Conformers **Author:** Frank Cai, Purdue University

The CIF structures underlying this study are drawn from the Cambridge Structural Database (CSD) and cannot be redistributed under the CSD licence. The manifest of refcodes and filter protocol used to select the corpus (3,501 candidate entries; formula, Z', R-factor, atom count, and space group per entry) is openly available at https://github.com/fronkt/symmc-flow/blob/master/data/csd_mol/manifest.csv and lets anyone with a licensed CSD installation regenerate the exact CIF corpus via `scripts/csd_export.py`. The factorized dataset (lattice, centroid, and orientation tensors derived from the CIFs) is itself derived from the licensed structures and is therefore available from the author on reasonable request, subject to the requester holding a CSD licence.

The SymMC-Flow implementation, the export and factorization pipeline, and the diagnostic and evaluation scripts are available at https://github.com/fronkt/symmc-flow and archived at https://doi.org/10.5281/zenodo.20822235.